1 Introduction

Section overview

- Both revenue recognition and accounting for inventories are affected by the application of accrual accounting.
- The key issue affecting revenue is the point at which an entity has fulfilled its performance obligations and is entitled to recognise revenue. This is when (or as) control of the good or service (ie, an asset) is passed to the customer.
- Control of the asset is described in IFRS 15, *Revenue from Contracts with Customers*, as the ability to direct the use of and obtain the remaining benefits of the asset.
- Inventories are covered by IAS 2, *Inventories*.
- The key issues affecting inventories include the following:
  - what costs may be included within the initial measurement of inventory; and
  - ensuring that inventories are carried at the lower of cost or net realisable value (as per IAS 2).

1.1 Background issues

Financial statements are prepared on the underlying assumption of the **accrual basis** of accounting, whereby effects of transactions are recognised **when they occur** and not when the cash associated with them is received or paid.

Therefore, when considering the recognition of revenue, this raises questions about **when** a transaction 'occurs':

- Is it when the buyer takes possession of the goods (is this when 'control' of the asset transfers)?
- Is it when services are provided in circumstances where the seller undertakes to come back to do additional work without charge if needed, eg, remedial work carried out by a building contractor?
- When does the revenue arise on a contract for the provision of services to a customer over time, such as under a maintenance contract of two years' duration? Only at the start of the contract, only in the middle of the contract, only at the end of the contract, or over the period of two years?

1.2 Context

Revenue is often the largest single figure in the financial statements. US studies have shown that over half of all financial statement frauds and requirements for restatements of previously published financial information involved revenue manipulation.

WorldCom was one of the world's largest telecommunication companies. In 2002 it was found that the directors had manipulated the financial statements using various means, including reallocating amounts from reserves and into revenue, in an attempt to overstate the earnings. Executive bonuses were tied to revenue targets and dubious accounting adjustments were constantly made to keep revenue 'on target'. By the time WorldCom filed for bankruptcy, revenue had been overstated by about $960 million.

In 2014, Tesco overstated its revenue with the resulting correction of profit, once the issue was uncovered, of over CU263 million. Key directors were accused of aggressive accounting policies with fines issued by the FRC and criminal action taken against members of the management team.

IFRS 15 sought to correct some of the judgmental areas by applying robust principles for the recognition and measurement of revenue. A consistent approach to revenue recognition is essential if financial statements are to faithfully represent the underlying economic reality of the entity. One of the main issues is the reporting of revenue within discrete periods, when under a single contract services are provided in the current and future periods. This allocation of revenue has a direct impact on the earnings for each period.

2 IFRS 15, *Revenue from Contracts with Customers*

Section overview

- The standard defines revenue as income arising in the course of an entity's ordinary activities.
- Revenue should be recognised to depict the transfer of goods or services (assets) to a customer in an amount that reflects the consideration to which the entity expects to be entitled.
- Revenue is recognised when (or as) the entity has transferred to the buyer control of the asset.
- IFRS 15 takes a standard approach to determining when revenue can be recognised and the amount that can be recognised. It is often referred to as a 'five-stage approach' but this term is not directly used in the standard.

2.1 Objective and scope

The objective of IFRS 15 is to establish the principles that should be applied to report on the nature, amount, timing and uncertainty of revenue and cash flows arising from a contract with a customer. (IFRS 15: para. 1)

IFRS 15 applies to all contracts with customers except:

- leases within the scope of IFRS 16, *Leases*;
- insurance contracts within the scope of IFRS 17, *Insurance Contracts* (outside of scope for this exam);
- financial instruments, or other contracts which are covered in IFRS 9, *Financial Instruments*, IFRS 10, *Consolidated Financial Statements*, IFRS 11, *Joint Arrangements*, IAS 27, *Separate Financial Statements* and IAS 28, *Investments in Associates and Joint Ventures*; and
- non-monetary exchanges between entities in the same line of business.

2.2 Revenue

Revenue is simply income arising in the course of an entity's ordinary activities and different terms may be used such as:

- sales
- turnover
- royalties

Dividend income and interest income is covered by IFRS Standards on financial instruments, not IFRS 15, and is presented as finance income rather than revenue.

Gains, such as those arising on the disposal of PPE, are not revenue as they do not arise in the course of ordinary activities. They are either presented separately as other income in the statement of profit or loss or are netted against the relevant expense category.

Definition

Revenue: Income arising in the course of an entity's ordinary activities. (IFRS 15: App. A)

2.3 Recognition

The core principle of IFRS 15 is that revenue is recognised to depict the **transfer of goods or services** to a customer in an amount that reflects the **consideration** to which the entity expects to be entitled in exchange for those goods or services (IFRS 15: para. 2).

The transfer of goods and services is evidenced by the transfer of **control**.

Revenue is recognised in accordance with this core principle by applying a standard approach as required by IFRS 15.

2.4 Five step model

Revenue is recognised by applying the following steps:

Step 1 Identify the contract with a customer.

Step 2 Identify the performance obligations in the contract.

Step 3 Determine the transaction price.

Step 4 Allocate the transaction price to the performance obligations in the contract.

Step 5 Recognise revenue when (or as) the entity satisfies the performance obligations.

Professional skills focus: Structuring problems and solutions

IFRS 15 suggests that this standard process is followed (applying the five steps stated above) in order to determine the most appropriate accounting treatment. Ensure that you learn and can apply these steps to a scenario. Application is important – you can't simply rote learn the steps, as it is important that you can identify which steps are relevant and apply them to the information provided. The interactive questions will help you to understand each stage and ensure that you have identified all the relevant information about the given scenario.

2.5 Identify the contract with a customer

The contract can be written, verbal or implied. For instance, in retail sales the contract is implied. Identifying the contract helps to establish the enforceable rights and obligations.

The criteria to be met are (IFRS 15: para. 9):

(a) All parties have **approved the contract**.

(b) The entity can identify **each party's rights** regarding the goods or services to be transferred.

(c) The **payment terms** can be identified.

(d) The contract has **commercial substance**.

(e) It is **probable that** the entity will collect the consideration to which it will be entitled in exchange for the goods or services.

A contract is **not** deemed to exist if either party can terminate the wholly unperformed agreement without having to compensate the remaining part.

2.6 Identify the performance obligations in the contract

A performance obligation is a promise to transfer to the customer either:

- a **distinct good or service** (or bundle of goods or services); or
- a **series of distinct goods or services** (IFRS 15: para. 22).

A good or service is distinct if:

- the customer can benefit from the good or service on its own or with other resources that are readily available; and
- the promise to transfer the good or service is separately identifiable from other promised in the contract.

If this does not apply, the good or service should be combined with other promised goods or services until a bundle of goods/services that is distinct can be identified.

Examples of performance obligations may include:

- sale or resale of goods or services
- acting as an agent to transfer goods or services
- granting licences
- granting options to purchase goods or services

Context example: Performance obligations

A software company, ClixPar is a reseller and installer of the popular software tool, Phage. It has a contract with a customer to provide the following:

- a software licence
- software installation
- any relevant software updates required in future years (up to three years)
- technical support for three years after installation

ClixPar sells each of these separately to other customers and other sellers also provide the same goods and services. The installation process is routine and ClixPar does not modify the software provided and the software remains functional without the updates and technical support.

So what are the performance obligations here?

The customer could benefit from the software using installation services from another supplier. It could equally use the software without updates or technical support, or by obtaining these from another supplier.

Therefore, each of the promises is distinct and there are four 'standalone' performance obligations in the contract.

If ClixPar were customising the software as part of the installation process to enable it to interface with the customer's other software applications, then the provision of software and its installation would not be distinct performance obligations, but would be combined to be a single performance obligation.

Interactive question 1: Identifying performance obligations

Skyward plc builds aircraft for the cargo and postal industries worldwide. It has been contracted to build three large cargo aircraft for Postal Europa Group (PEG). Skyward plc is to build the aircraft and is contracted to deliver them to PEG as each one is completed. It is expected to take six months for the construction of each aircraft, with the total time required therefore being 18 months.

Requirement

Identify the performance obligation(s) in the contract. See **Answer** at the end of this chapter.

2.7 Determine the transaction price

This is the amount of consideration in a contract to which an entity expects to be entitled for transferring promised goods or services to a customer (IFRS 15: para. 47).

This may include fixed and variable elements but not amounts collected on behalf of third parties. To determine the transaction price, the entity must consider:

- variable consideration
- the existence of a significant financing component
- non-cash consideration

2.7.1 Variable consideration

Definition

Variable consideration: Promised consideration may include a variable amount if the contract contains discounts, rebates, performance bonuses, penalties etc.

An entity shall estimate the amount of the variable consideration in a transaction price using whichever of the following methods better predicts the amount:

- the **expected value** – the sum of the probability-weighted amounts in a range of different outcomes; or
- the **most likely amount** – the single most likely amount in a range of different outcomes (IFRS 15, para 53).

Variable consideration is only included in the transaction price to the extent that it is **highly probable** that there will not be a significant reversal of revenue when the uncertainty associated with the variable consideration is resolved (IFRS 15: para. 56). Any changes to the variable consideration are allocated to all of the performance obligations in the contract. It should be noted that changes to variable consideration are outside of the scope of the *Financial Accounting and Reporting* syllabus, however, you should be aware that this is covered in your *Corporate Reporting* studies at Advanced level.

2.7.2 Significant financing component

Definition

Significant financing component: A significant financing component exists where the customer receives a benefit from financing due to the consideration being transferred after the satisfaction of the performance obligation (IFRS 15, para. 60). This may be the case regardless of whether the financing is implicit or explicit in the contract.

In such a case the consideration should be discounted to reflect the time value of money. Revenue should be recognised based on the price at which the transaction would be settled on a cash basis (the cash price). The significant financing component is the difference between the cash price and the amount that will be paid by the customer. The significant financing component is separately accounted for as finance income over the period between the date of the transaction and the date the full consideration is received.

Note: It is also possible for the entity to receive benefit from a significant financing component if it receives consideration in advance of the performance obligations being satisfied. You will not face this situation in the *Financial Accounting and Reporting* exam.

2.7.3 Non-cash consideration

Definition

Non-cash consideration: Consideration which is not in the form of cash, such as other assets, products or shares.

An entity may accept payment for the transfer of goods or services in a form other than cash. In this case, the non-cash consideration should be measured at its fair value at the date of transfer (IFRS 15, para. 66). If the fair value of the non-cash consideration cannot be determined, the entity should measure the transaction at the stand-alone selling price of the goods or services transferred to the customer.

Worked example: Significant financing component

A car retailer offers its customers a choice of paying for a car at its cash price today, 1 January 20X7, of CU17,150, or to pay a nothing today and make a single payment of CU20,000 in two years' time. The car retailer has calculated a finance cost of 8% applies to the transaction. A customer purchased a car on 1 January 20X7 and chose to pay CU20,000 on 1 January 20X9.

Requirement

How is the sale of the car accounted for in the year ended 31 December 20X7?

Solution

	CU
Revenue recognised	17,150
Finance income (W)	1,372
Carrying amount of receivable (W)	18,522

The revenue recognised is based on the price of the car had the transaction been settled immediately in cash.

The difference between this price and the amount receivable on 1 January 20X9 represents finance income which is recognised across the two-year period before payment is made.

WORKING

Finance income and receivable

The finance income for the year and carrying amount of the receivable at 31 December 20X7 are calculated as follows:

Year	Opening (CU)	Finance income at 8%	Closing (CU)
1	17,150	1,372	18,522
2	18,522	1,478*	20,000

* Adjusted for rounding differences

Context example: Interest-free credit

Comfy Couches Ltd delivers an item of furniture to a customer on 1 September 20X7 for CU2,500, with a one-year interest-free credit period. The company normally applies an interest rate of 7% to the sale of furniture.

Despite the fact that the customer receives an interest-free credit period, Comfy Couches Ltd must calculate the fair value of the sale by applying its standard interest rate of 7% to the transaction.

The fair value of the furniture is therefore CU2,500/1.07 = CU2,336, which is the amount that should be recognised as revenue on the sale.

- Interest revenue of CU164 (2,500 – 2,336), which would be recognised over the credit period
- Sales revenue of CU2,336, which would be recognised on 1 September 20X7 when control of the furniture passes to the customer

2.8 Allocate the transaction price to the performance obligations in the contract

The transaction price is allocated to each performance obligation on the basis of the **stand-alone selling price** of each distinct good or service in the contract (IFRS 15: para. 74). If the good or service does not have a directly observable stand-alone selling price, it will need to be estimated.

A contract may be for the supply of a bundle of goods or services, often at a discount to the stand- alone selling prices. An example of this is a mobile phone service contract which includes a free handset. The contract should be unbundled and in this case two performance obligations can be identified, being the provision of the handset and the provision of the service plan. In accordance with IFRS 15 the entity should allocate the total transaction price between the handset and service plan, based on their stand-alone selling prices.

Context example: Unbundling

A mobile phone company gives customers a free handset when they sign a two-year contract for the provision of network services. The handset has a stand-alone price of CU800 and the contract is for CU60 per month.

Under IFRS 15, there are two performance obligations: the provision of the handset to the customer, which is satisfied when control of the handset transfers to the customer, and provision of the network services over the two years. Some revenue must be recognised on delivery of the handset to the customer because this constitutes satisfaction of the first performance obligation. The remainder is recognised as network services are provided. The total transaction price is allocated between the performance obligations as follows:

	CU	%
Handset	800	36
Contract – Two years (CU60 × 24 months)	1,440	64
Total value	2,240	100

CHAPTER 6

The total receipts are CU1,440, which is the amount which must be allocated to the separate performance obligations. Revenue will be recognised as follows:

	CU
Year 1	
Handset (1,440 × 36%)	518
Contract (1,440 – 518)/2	461
	979
Year 2	
Contract as above	461

2.9 Recognise revenue when (or as) a performance obligation is satisfied

A performance obligation is satisfied when control of the good or service specified in the contract is transferred to the customer (IFRS 15: para. 31).

A performance obligation can be satisfied **at a** point in time, such as in retail sales, or **over time**, such as a construction contract taking place over weeks, months or even years.

2.9.1 Performance obligations satisfied at a point in time

Where a performance obligation is satisfied **at a point in time**, that point will occur when **control is transferred**. At that point the customer is able to direct the use of the asset and obtain substantially all the remaining benefits from it (IFRS 15: para. 33).

The following events can indicate that control has been transferred (IFRS 15: para. 38):

- The entity has a present right to payment for the asset.
- The customer has legal title to the asset.
- The entity has transferred physical possession of the asset.
- The significant risks and rewards of ownership have been transferred to the customer.
- The customer has accepted the asset.

2.9.2 Performance obligations satisfied over time

In order to be considered satisfied over time, a performance obligation must meet one of the following criteria (IFRS 15: para. 35):

- The customer simultaneously receives and consumes the benefits as the entity performs.
- The entity's performance creates or enhances an asset that the customer controls as the asset is created or enhanced.
- The entity's performance does not create an asset with an alternative use to the entity and the entity has an enforceable right to payment for performance completed to date.

Where performance obligations are satisfied over time, the entity recognises revenue by measuring progress towards complete satisfaction of the performance obligation. Progress can be measured using two methods:

- the output method – measuring the value to the customer of goods or services transferred to date; or
- the input method – measuring the cost to the entity of goods or services transferred to date (IFRS 15: para. B14).

In the early stages of a contract, it may not be possible to reasonably measure the outcome of a performance obligation, but the entity is entitled to recover costs incurred. In this case, revenue can be measured to the extent of costs incurred.

Note: IFRS 15 does not explicitly cover the accounting treatment for contracts that are expected to be loss-making. These instead fall under IAS 37, *Provisions, Contingent Liabilities and Contingent Assets* which is covered in Chapter 9.

2.9.3 Contract assets, contract liabilities and trade receivables

Contracts with customers will be presented in an entity's statement of financial position as a contract liability, a contract asset or a receivable, depending on the relationship between the entity's performance and the customer's payment. (IFRS 15: para. 105)

A contract liability is recognised and presented in the statement of financial position where a customer has paid an amount of consideration prior to the entity fulfilling its performance obligations by transferring control of the related good or service to the customer. (IFRS 15: para. 106)

When the entity has satisfied its performance obligations but the customer has not yet paid the related consideration, this will give rise to either a contract asset or a trade receivable. A contract asset is recognised when the entity's right to consideration is conditional on something other than the passage of time; for instance, future performance. A receivable is recognised when the entity's right to consideration is unconditional except for the passage of time. (IFRS 15: para. 107)

Where revenue has been invoiced but the customer has not yet settled the invoice, a trade receivable is recognised.

Professional skills focus: Applying judgement

A professional accountant will be required to apply judgement when measuring progress towards satisfaction of performance obligations. They must determine which method to use and how to appropriately measure the value or cost transferred under each method.

Worked example: Performance obligation satisfied over time

Associated Solutions Ltd is creating a bespoke software system for an insurance company. The software will transfer to the customer on completion of the contract and Associated Solutions Ltd has no alternative use for the software. The contract started on 1 January 20X7 with an estimated completion date of 31 December 20X9. Associated Solutions Ltd has an enforceable right to receive the payment for work completed to date based on a total contract price of CU2 million. Associated Solutions Ltd expects to earn a profit on the contract. The company uses the input method to determine completion of the contract based on staff time incurred to date.

(1) Costs incurred in the period to 31 December 20X7 amounted to CU800,000.

(2) Company time sheets indicate that half of the budgeted staff time for the contract has been incurred to date.

(3) Associated Solutions Ltd has invoiced CU400,000 to customers, of which half has been paid by the customer at 31 December 20X7.

Requirements

1 Explain and show the amount of revenue and costs that Associated Solutions Ltd can recognise in relation to the contract in its statement of profit or loss for the year ended 31 December 20X7.

2 What amounts should be presented in the statement of financial position in respect of the contract at 31 December 20X7?

Solution

1 This is a contract in which the performance obligation is satisfied **over time**. The entity is carrying out the work for the benefit of the customer – it does not have an alternative use for the asset – and it has an enforceable right to payment for work completed to date.

Associated Solutions Ltd uses the input method to measure progress towards satisfaction of the performance obligation, based on the proportion of staff time incurred compared to the total expected staff time required. As 50% of the total expected staff time has been incurred in the year to 31 December 20X7, the performance obligation is considered to be 50% complete.
Associated Solutions Ltd can therefore recognise revenue of CU1 million (50% x CU2 million).

Costs associated with the contract that can be recognised in the year to 31 December 20X7 are the CU800,000 incurred in the year.

2 Associated Solutions Ltd should present a contract asset of CU600,000 based on the difference between the revenue recognised of CU1,000,000 and the amount invoiced to date of CU400,000. This represents the entity's right to future consideration in respect of work completed to date.

There is a trade receivable of CU200,000, which is the amount invoiced for which payment has not yet been received.

Interactive question 2: Advance sales

A DIY store is about to sell a new type of drill. Customer demand is high and the store has taken advance orders for the drill. The selling price of the drill will be CU50 and so far two hundred customers have paid an initial 10% deposit on the selling price of the drill. No drills are yet held in inventory.

Requirement

What amount should be recognised as revenue? See **Answer** at the end of this chapter.

Interactive question 3: Rendering of services

DataUs Ltd entered into a CU210,000 fixed-price contract for the provision of data analysis services for its customer, a sustainable fashion retailer. DataUs Ltd satisfies its performance obligations over time and at the end of 20X7, its first accounting period, costs of CU45,000 have been incurred in performing the work and CU40,000 of these are recoverable in full from the customer. DataUs Ltd measures progress towards complete satisfaction of the performance obligation using the input method based on costs incurred to date as a proportion of total expected costs.

Requirements performance 45/90 = 0.5

Calculate the revenue to be recognised in 20X7 on the alternative assumptions that:

(a) The costs to complete can be reliably estimated at CU90,000.

(b) The costs to complete cannot be reliably estimated.

See **Answer** at the end of this chapter.

Interactive question 4: Service contract

An entity entered into a contract for the provision of services over a two-year period. The total contract price was CU150,000. In the first year, costs of CU60,000 were incurred; the contract did not progress as expected and at the end of the first-year management was not sure of the total expected costs of the project, but believed that the costs incurred to date would be recovered from the customer.

Requirement

What amount of revenue should be recognised for the first year of the contract?

See **Answer** at the end of this chapter.

Interactive question 5: Goods and services

An entity sells an item of equipment to a customer on 1 January 20X7 for CU1.5 million, the goods are transferred and control of that equipment is taken by the customer immediately. Due to the specialised nature of the equipment the entity has agreed to provide free support services for the next two years, despite the cost to the entity of that support being estimated at CU120,000 in total. The entity usually earns a gross margin of 20% on such support service contracts.

Requirement

How much revenue should the entity recognise for the year ended 30 April 20X7?

See **Answer** at the end of this chapter.

Interactive question 6: Servicing fees

On the last day of its current accounting period, Computer Ltd completes the handover of a new system to its customer and raises an invoice for CU800,000. This price includes free after-sales support for the next two years, which costs CU35,000 each year. Computer Ltd normally earns a gross profit margin of 17.5% on such support activity.

Requirement

Calculate the revenue to be included in Computer Ltd's current year statement of profit or loss in respect of this sale.

	CU
Standalone price of after-sales support Standalone price of system	
Total	
Transaction price allocated to the provision of the system	

The transaction price allocated to the provision of the system is recognised as revenue in the year.

See **Answer** at the end of this chapter.

2.10 Application of IFRS 15

The Application Guidance of IFRS 15 has a number of additional provisions from the Standard. Some of the specific situations are relevant to your *Financial Accounting and Reporting* studies.

2.10.1 Principal versus agent

IFRS 15 specifically excludes from the transaction price attached to a contract amounts collected on behalf of third parties (IFRS 15: para. 47). This applies to an agent collecting amounts on behalf of a principal.

Individually or in combination, the following criteria indicate that the entity is acting as a principal (IFRS 15: para. B37):

- The entity is primarily responsible for fulfilling the promise to provide the specified good or service to the customer.
- The entity has inventory risk before the specified good or service has been transferred to the customer or after transfer of control to the customer.
- The entity has discretion in establishing prices for the goods or services.

An entity is a principal if it controls the promised good or service before it is transferred to the customer. In this case, when the performance obligation is satisfied the entity recognises the revenue.

An entity is an agent if its performance obligation is to arrange for the provision of goods or services by another party. When this performance obligation is satisfied, it recognises revenue only for the fee or commission to which it is entitled (IFRS 15: para. B36).

Worked example: Agent or principal?

An entity runs a website which enables customers to buy goods from a range of suppliers. Prices are set by suppliers and payments are processed through the entity's website. Customers pay in advance and goods are delivered directly from the supplier to the customer.

The entity receives a commission of 10% of the sales price and has no further obligation to the customer after arranging for the products to be shipped.

Requirement

Is the entity an agent or the principal?

Solution

The entity is acting as an agent based on the following points:

- Goods travel directly from the supplier to the customer, so the entity never has physical custody of them and does not bear the associated risk.
- The supplier, not the entity, has the obligation to the customer.
- The entity does not set prices.

The entity should only recognise as revenue the commission received from suppliers.

2.10.2 Extended warranties

It is necessary to distinguish between warranties which give the customer assurance that an item will function as intended (standard warranties which are not separately paid for by the customer) and warranties which provide the customer with an additional distinct service (such as free repairs over a specified period). Warranties may be offered as part of a legal obligation (through a contract) or through a constructive obligation (such as a department store's 'no quibble' return policy within six months of purchase).

A warranty which does not promise an additional service, but simply provides assurance that an item will work as intended is accounted for in accordance with IAS 37, *Provisions, Contingent Liabilities and Contingent Assets* (IFRS 15: para. B28–30).

A warranty which provides the customer with an additional service is within the scope of IFRS 15. This is regardless of whether the customer pays for the additional service warranty or not. This is often referred to as extended or additional warranties and are usually paid for and go 'above and beyond' the usual level of warranty. Sometimes this may be paid for, or a company may decide to offer this as an incentive to purchase the item.

A warranty which the customer purchases separately will always be an additional service warranty, within the scope of IFRS 15. Such warranties represent separate performance obligations and therefore the revenue attributable to the warranty must be recognised separately from the revenue from the sale of goods. Remember, this is an additional service agreed between the seller and the purchaser, and therefore there is a **contract** which requires treatment under IFRS 15.

An additional service warranty is accounted for as a separate performance obligation within a contract and a portion of the transaction price is allocated to it.

Worked example: Warranties

Kaptcha Ltd sells a machine to Cylon Ltd for CU490,000 (the list price of the machine). A warranty is included in the sales contract providing assurance that the machine will operate as expected for one year from the date of purchase.

Cylon Ltd is a new customer, so Kaptcha Ltd also provides an additional extended warranty (at no extra cost to Cylon Ltd) which guarantees that the machine will operate as expected for a further twelve month period after the standard warranty has elapsed.

Extended twelve-month warranties are normally charged at CU10,000.

Requirement

Explain how the two warranties and the sale of the machine should be accounted for (ignore the effect of any discounting).

Solution

Standard warranty

The warranty that provides a guarantee that the machine complies with the specifications agreed in the contract and will operate as promised for one year from the date of purchase is a standard warranty at no cost to Cylon Ltd.

Therefore, it should be accounted for in accordance with IAS 37. This will be explained in further detail in Chapter 9.

Additional warranty

The additional warranty provides an additional performance guarantee for a further twelve months after the standard warranty expires. This second warranty is an additional service warranty and should be treated as a separate performance obligation. Revenue will be recognised when the performance obligation is satisfied which will be when the additional warranty period takes place.

Sale of the machine

The transaction price of CU490,000 should be allocated to the two performance obligations in accordance with their standalone selling prices:

Performance obligation	Stand-alone selling price CU	% of total	Transaction price allocated CU
Machine	490,000	98	480,200
Extended warranty	10,000	2	9,800
Total	500,000	100	490,000

Revenue of CU480,200 from the sale of the machine will be recognised when control of the machine is transferred to the customer (for example, upon delivery to Cylon Ltd). Revenue of CU9,800 in respect of the additional warranty will be recognised as the warranty period takes place (over the 24-month period of the warranty).

Professional skills focus: Assimilating and using information

It is important to recognise that warranties may be dealt with in two different ways, under IAS 37 as a provision for future costs (such as in a standard warranty) or under IFRS 15 (as revenue) where the warranty has a performance obligation attached to it. Read the question and the scenario carefully to ensure that you have fully understood the issues. You need to be able to evaluate the relevance of the information that is being provided and identify the critical facts to account for the transactions correctly.

2.10.3 Sale with a right of return

Some contracts give the customer the right to return the product and receive a refund, a credit or a replacement.

The entity should recognise (IFRS 15: para. B21):

- revenue for the transferred products
- a refund liability
- an asset in respect of products to be returned

Worked example: Sale with a right of return

Wholesaler Ltd supplies popcorn machines to retailers across Bangladesh. It only supplies wholesale to department stores and other independent retailers. The popcorn machines are sold to retailers for
CU10.00 each, with each popcorn machine costing CU5.00 to manufacture.

Wholesaler Ltd estimates that 8% of its popcorn machines will be returned by customers.

Wholesaler Ltd has a contract term with all of its customers to allow any returns for a full refund within three months of the date of sale.

On 1 April 20X8, Wholesaler Ltd sold 500 machines to one of its largest customers, the popular home store Likeland.

Requirement

How should the revenue be recognised?

Solution

When the popcorn machines are sold revenue is recognised in respect of machines that are not expected to be returned, and a liability is recognised in respect of the machines that are expected to be returned:

Revenue

	CU
500 units × CU10 per unit	5,000
Less: proportion expected to be returned	
8% × 500 × CU10	(400)
Revenue recognised on 1 April	4,600

This is recognised by:

	CU	CU
DR Bank/receivable	5,000	
CR Revenue		4,600
CR Refund liability		400

In addition, the machines transferred to customers are recognised in cost of sales to the extent that they are not expected to be returned and an asset is recognised in respect of the machines that are expected to be returned:

Cost of sales

	CU
500 × CU5	2,500
Less: proportion expected to be returned	
8% × 500 × CU5	(200)
Cost of sales recognised on 1 April	2,300

This is recognised by:

	CU	CU
DR Asset	200	
DR Cost of sales	2,300	
CR Inventories		2,500

The CU200 asset and CU400 refund liability may be reversed once the three-month period has expired.

2.10.4 Repurchase agreements

Under a repurchase agreement an entity sells an asset and has an option to repurchase it. This can come in one of three forms (IFRS 15: para. B64):

(a) An obligation to repurchase (a forward contract)

(b) A right to repurchase (a call option)

(c) An obligation to repurchase at the customer's request (a put option)

Under a **forward contract** or a **call option** the customer does not obtain control of the asset because that control is limited by the repurchase option. The contract is accounted for as a lease if the repurchase price is below the original selling price, or a financing arrangement if the repurchase price is equal to or above the original selling price (IFRS 15: para. B66).

Under a **put option**, if the customer does not have sufficient economic incentive to exercise the right to request repurchase, the agreement should be treated as if it were a sale with a right of return. If the customer **does** have sufficient economic incentive to exercise the option, the entity should account for the agreement as a lease in accordance with IFRS 16. If the **repurchase price is greater than or equal to the original selling price** and is above the expected market value of the option, the contract is treated as a financing arrangement.

Worked example: Repurchase agreement

Vass Ltd, a property development company, sold a property to another entity for CU4 million when the fair value of the property was CU5 million. Further investigation uncovered an agreement whereby Vass Ltd could repurchase the property after one year for CU4.32 million.

Requirement

How should this transaction be accounted for?

Solution

Vass Ltd has a right to repurchase the property – a call option. The repurchase price is above the original selling price, so this is in effect a financing arrangement.

The sale of the property at 20% below fair value is sufficient to cast doubt on whether a real sale has been made. Also, the repurchase price is below fair value at the date of sale and represents a return to the financial institution of 8% ((CU4.32m – 4m) as a percentage of CU4 million) on the amount paid out.

The substance of the arrangement appears to be that other entity has granted Vass Ltd a one year loan secured on the property, charging interest at 8%.

IFRS 15 reflects substance and requires that the transaction should be accounted for by Vass Ltd by:

- continuing to recognise the property as an asset;
- crediting the CU4 million received to a financial liability account;
- recognising CU0.32 million as a finance cost in profit or loss over the year and crediting it to the liability account; and
- derecognising the liability when the CU4.32 million cash is paid out.

2.10.5 Further application guidance

IFRS 15 Appendix B also provides specific application guidance in respect of the following:

Consignment sales	Under such arrangements, the buyer of the goods undertakes to sell them on, but on behalf of the original seller. Usually the buyer has not obtained control of the goods. This may be evidenced by the fact that the seller can require the return of the goods or transfer them to another dealer. In this case the original seller only recognises a sale **when the buyer sells the goods on to a third party**. (IFRS 15: para. B.77)

Bill-and-hold arrangements	Under these arrangements, an entity bills the customer for the product but delivery is delayed with the agreement of the customer, perhaps because the customer is short of space. In this case the entity must determine the point in time at which it has satisfied its performance obligation by transferring control of the product to the customer. It may be that the customer is still able to exercise control without having physical possession of the product. (IFRS 15: para. B.79)
Licensing	The grant of a licence may form a separate performance obligation within a contract or, if the promise to grant the licence is not distinct from the other promised goods or services in the contract, all the promises are treated as a single performance obligation. Where the granting of a licence is a separate performance obligation, the grantor must determine, on the basis of the contract, whether the licence transfers to the customer over time or at a point in time. A licence to access intellectual property as it existed when the licence was granted is a point in time transfer and results in the related revenue being recognised at the grant date; a licence to access intellectual property as it exists throughout a licence period (ie, reflecting updates) is an over-time transfer and results in the related revenue being recognised over the licence period. (IFRS 15: para. B.56)

2.11 Disclosure

The following amounts should be disclosed unless they have been presented separately in the financial statements in accordance with other standards (IFRS 15: para. 110):

- revenue recognised from contracts with customers, disclosed separately from other sources of revenue;
- any impairment losses recognised (in accordance with IFRS 9) on any receivables or contract assets arising from an entity's contracts with customers, disclosed separately from other impairment losses;
- the opening and closing balances of receivables, contract assets and contract liabilities from contracts with customers;
- revenue recognised in the reporting period that was included in the contract liability balance at the beginning of the period; and
- Revenue recognised in the reporting period from performance obligations satisfied in previous periods (such as changes in transaction price).

3 IAS 2, *Inventories*

Section overview

- Inventories should be measured at the lower of cost and net realisable value (NRV).
- Cost comprises the costs of:
  - purchase
  - conversion
  - bringing the inventories to their present location and condition

- The comparison of cost and NRV should be performed on an item-by-item basis although similar items may be grouped together.

3.1 Objective and scope

The objective of IAS 2, *Inventories* is to prescribe the accounting treatment for inventories. In particular it provides guidance on the **determination of cost** and its subsequent recognition as an expense, including any write-down to **net realisable value**.

IAS 2 applies to all inventories except the following:

- financial instruments (eg, shares, bonds) (see Chapter 8)
- biological assets (covered in IAS 41, *Agriculture)*

Biological assets are outside the scope of the *Financial Accounting and Reporting* syllabus.

Certain inventories are exempt from the standard's measurement rules ie, those held by:

- producers of agricultural, forest and mineral products; and
- commodity-broker traders.

3.2 Revision of *Accounting* material

IAS 2 is covered in some detail in the *Accounting* syllabus, so this section is mainly revision.

Definition

Inventories: Assets which are:

- held for sale in the ordinary course of business;
- in the process of production for such sale; or
- in the form of materials or supplies to be consumed in the production process or in the rendering of services.

Inventories can include:

- **goods purchased and held for resale**
- **finished goods**
- **work in progress** being produced
- **raw materials** awaiting use

3.2.1 Measurement of inventories

Inventories should be measured at the **lower of cost and net realisable value (NRV).**

Definitions

Inventory cost: Comprises all costs of purchase, costs of conversion and other costs incurred in bringing the inventories to their present location and condition.

Net realisable value: The estimated selling price in the ordinary course of business less the estimated costs of completion and the estimated costs necessary to make the sale.

A write-down of inventories from cost to NRV would normally take place on an **item-by-item basis**, but similar or related items may be **grouped together**. This grouping is acceptable for, say, items in the same product line, but it is not acceptable to write-down inventories based on a whole classification (eg, finished goods) or a whole business.

3.2.2 Cost of purchase

Definitions

Cost of purchase: IAS 2 lists the following as comprising the costs of purchase of inventories:

- purchase price; plus
- import duties and other non-refundable taxes; plus
- transport, handing and other costs which are directly attributable to the acquisition of finished goods, services and materials; less
- trade discounts, rebates and other similar amounts.

Costs of conversion: Costs that are specifically attributable to units of production and a systematic allocation of fixed and variable production overheads incurred in converting raw materials into finished goods.

Costs of conversion of inventories consist of two main parts:

- costs **directly related** to the units of production eg, direct materials, direct labour; and
- fixed and variable **production overheads** that are incurred in converting materials into finished goods.

The allocation of fixed production overheads to the costs of conversion is based on **the normal capacity of the production facilities.**

Definitions

Fixed production overheads: Those indirect costs of production that remain relatively constant regardless of the volume of production, such as depreciation and maintenance of factory buildings and equipment, and the cost of factory management and administration.

Variable production overheads: Those indirect costs of production that vary directly, or nearly directly, with the volume of production, such as indirect materials and labour. (IAS 2, para.12)

IAS 2 emphasises that fixed production overheads must be allocated to items of inventory on the basis of the **normal capacity of the production facilities**. This is an important point.

- **Normal capacity** is the expected achievable production based on the average over several periods/seasons, under normal circumstances.
- The above figure should take account of the capacity lost through **planned maintenance**.
- If it approximates to the normal capacity then the **actual level of production** can be used.
- The allocation of variable production overheads to each unit is based on the **actual use** of production facilities.

As a result:

- **Low production** or **idle plant** will **not** result in a higher fixed overhead allocation to each unit.
- **Unallocated overheads** must be recognised as an expense in the period in which they were incurred.

- When production is **abnormally high**, the fixed production overhead allocated to each unit will be reduced, so avoiding inventories being stated at more than cost.

Other costs

Any other costs should only be recognised if they are incurred in bringing the inventories to their **present location and condition.**

IAS 2 lists types of cost that **would not be included** in cost of inventories. Instead, they should be recognised as an **expense** in the period in which they are incurred.

These include:

- **abnormal amounts** of wasted materials, labour or other production costs;
- **storage costs** (except costs that are necessary in the production process before a further production stage);
- **administrative overheads** not incurred to bring inventories to their present location and condition; and
- **selling costs**.

Interactive question 7: Fixed production overheads

A business plans for fixed production overheads of CU50,000 and annual production of 100,000 units in its financial year.

A fire at the factory results in production being only 75,000 units, with no saving in fixed production overheads.

Requirement

At which amount per unit should the fixed production costs be allocated to inventory before and after the fire? How should any unallocated overheads be accounted for?

See **Answer** at the end of this chapter.

Interactive question 8: Cost of inventories

A manufacturing business incurs various types of expenditure. The directors of the business would like to include all of the expenditure in the cost of inventory.

Requirement

Using the proforma provided, identify the expenditures to be included in the cost of inventories and those to be recognised as an expense as incurred.

Notes

1. In terms of the normal operating cycle of a business, all costs up to the time goods are taken into inventory will be costs incurred in bringing items **to their present location and condition**. But all costs of holding goods in inventory, selling the goods and collecting outstanding receivables are not incurred for this reason; nor are the general costs of accounting for and managing the business.
2. Fixed production costs/overheads are to be included in inventory values, but only at the rates based upon **normal levels of output**. These rates should not be increased as a result of production being below expected levels, as a result of plant failures, for example.

	Include in cost of inventories	Recognised as an expense as incurred
Supplier's gross price for raw materials		
Quantity discounts allowed by supplier		
Purchase taxes and duties charged by supplier and recoverable from taxing authorities		
Costs of transporting materials to the business' premises		
Labour costs directly incurred in the processing of raw materials		
Variable costs, such as power, incurred in the processing of raw materials		
Fixed production costs/overheads, such as rent for the processing factory and depreciation charges on the plant used in the processing		
Costs of holding finished goods in inventory		
Costs of transporting goods to customer on sale		
Purchase taxes charged to customer on sale		
Commission payable to salesmen on sale of the goods		
Allowance for irrecoverable debts in relation to trade receivables		
Costs of accounts department		
Head office costs relating to the overall management of the business		

See **Answer** at the end of this chapter.

Techniques for the measurement of cost

Two techniques are mentioned by the standard, both of which produce results that **approximate to cost**, and so both of which may be used for convenience.

(a) **Standard costs**: These are set up to take account of normal levels of raw materials used, labour time etc. They are reviewed and revised on a regular basis.

(b) **Retail method**: This is often used in the retail industry where there is a large turnover of inventory items, which nevertheless have similar profit margins. The only practical method of inventory valuation may be to take the total selling price of inventories and deduct an overall average profit margin, thus reducing the value to an approximation of cost. The percentage will take account of reduced price lines. Sometimes different percentages are applied on a departmental basis.

Interactive question 9: Retail method

A retailer identifies inventories at the end of an accounting period as follows:

- Department A: inventories with a selling price of CU30,000. This department makes a 25% gross profit on its sales.

- Department B: inventories with a selling price of CU21,000. This department sets its selling prices at cost plus 50%.

Requirement

Calculate the cost of inventories in each department. See **Answer** at the end of this chapter.

3.2.3 Cost formulae

It is possible to attribute specific costs to items that are not interchangeable and to items produced for specific projects or customers and it is these costs which are used in arriving at inventory valuations.

But **many inventories include items that are interchangeable** with each other, in which case **it is not possible to identify a specific cost for a specific item**. In these cases, **cost formulae** should be used, which make assumptions about which of the items produced have been sold and which are still held in inventory, and therefore about the cost of inventory.

Only two cost formulae are allowed under IAS 2:

First-in, first-out (FIFO)	Weighted average cost
This assumes a **physical flow of items** whereby **those produced earliest are the first to be sold**. The items produced most recently are the ones in inventory, to be measured at the most recent production cost.	This formula calculates an **average cost of production** (either at the end of each period or after each new batch has been produced, depending on the circumstances of the company) and measures inventories at that average cost.

Notes

1. The last-in, first-out (LIFO) formula (which makes an assumption about the physical flows of items that is the opposite of FIFO) is not permitted by IAS 2. The reasoning, not included in the standard, is that LIFO is not a reliable representation of the actual flow of items into and out of inventory.
2. The same cost formula must be used for all inventories having a similar nature. This limitation on management choice is aimed to ensure that like items are accounted for in like ways.

3.3 Net realisable value (NRV)

As a general rule assets should not be carried at amounts **greater than those to be realised from their sale or use**. This applies to inventory **where NRV falls below cost**. There are a number of reasons why this may be the case, including the following:

- **an increase in costs or a fall in selling price**
- a **physical deterioration** in the condition of inventory
- **obsolescence** of products
- a **strategic decision** to manufacture and sell products at a loss
- **errors** in production or purchasing

Where NRV falls below cost the inventory is written down to its recoverable amount and **the fall in value is charged to profit or loss**. The write-down may be of such size, incidence or nature that it must be **disclosed separately**.

Notes

1. In the case of **incomplete items**, NRV must take account of **costs to complete**.
2. In the absence of a contractually agreed selling price, the **best estimate** must be made of the likely selling price and then appropriate deductions made from it.
3. Materials to be incorporated into a finished product should only be written down **if that finished product will be sold at below its cost**.
4. Net realisable value must be **reassessed at the end of each period** and compared again with cost. This may result in the **reversal** of all or part of the original write-down.

3.4 Recognition as an expense

Once an item has been sold, it cannot remain in inventories as it no longer meets the *Conceptual Framework* definition of an asset. Its carrying amount is recognised as an **expense** in the accounting period in which the item is sold and the **related revenue recognised**.

3.5 Disclosure

The financial statements should disclose the following:

- **accounting policies** adopted in measuring inventories, including the cost formula used;
- **total carrying amount of inventories** and the carrying amount in classifications appropriate to the entity (eg, merchandise, production supplies, materials, work in progress, finished goods);
- **carrying amount** of inventories carried at fair value less costs to sell;
- the amount of inventories **recognised as an expense** in the period;
- the amount of any **write-down** of inventories **recognised as an expense** in the period;
- the amount of any **reversal of any write-down** that is recognised as a reduction in the amount of inventories recognised as an expense in the period;
- **circumstances or events** that led to the reversal of a write-down of inventories; and
- carrying amount of inventories **pledged as security for liabilities**.

The financial statements must also disclose one of two things:

- the **cost of inventories** recognised as an expense during the period; or
- the **operating costs**, applicable to revenues, recognised as an expense during the period, classified by their nature.

4 Ethical and judgement issues

Section overview

Professional accountants should apply scepticism when assessing management's assumptions when recognising revenue. Revenue is one of the most significant balances in the financial statements.

One of the most common ways to increase profit is to **overstate revenue**. Both the timing of revenue recognition and its measurement are crucial, as these both require judgement. For example, identification of a number of performance obligations within a contract may result in different timing of revenue recognition; variable consideration may increase or decrease transaction price. Increased profit in turn increases distributable reserves and therefore allows the payment of larger dividends and often triggers the payment of executive bonuses. In addition, the advancement of revenue and hence inflated profit for a particular period may be beneficial if the company is looking to expand or attract new owners, such as before a floatation of shares.

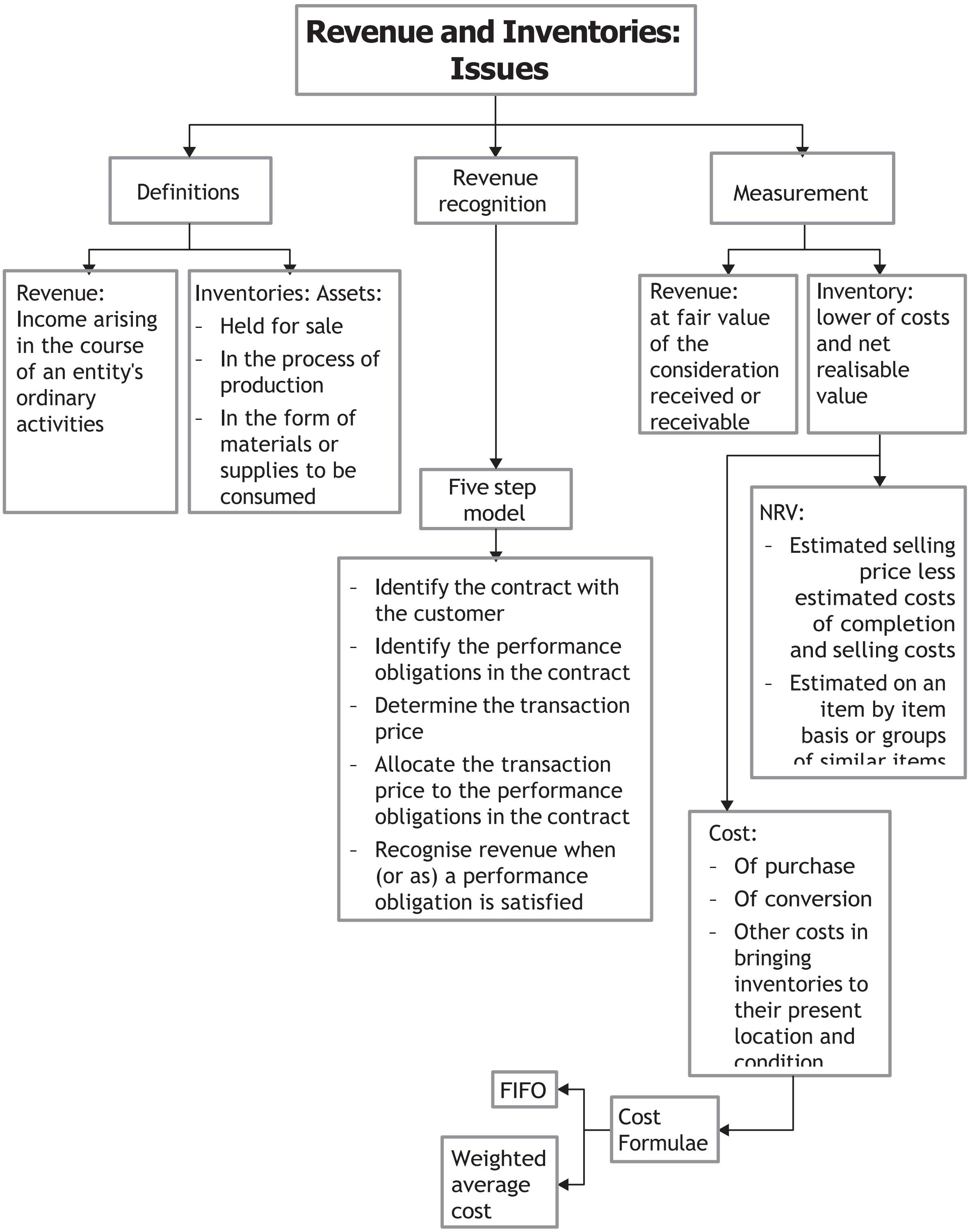
Because revenue has such a direct impact on earnings and because many investors focus on revenue growth as an important measure of performance, management may well be tempted to be very optimistic in its judgements so that the amount of revenue recognised is maximised. Professional accountants need to be very clear about the need to ensure that there are robust internal policies for the recognition of revenue which are in line with IFRS 15 and that they are applied consistently from one period to another.

Professional skills focus: Applying judgement

Be aware of potential ethical issues in your exam, such as aggressive accounting to overstate revenue for the period or the incorrect inclusion of expenses within the valuation of inventory at year end.

Consider how these practices may affect the fundamental ethical principles expected of a professional accountant.

Summary


Revenue and Inventories: Issues
Definitions
Revenue recognition
Measurement
Revenue: Income arising in the course of an entity's ordinary activities
Inventories: Assets:
- Held for sale
- In the process of production
- In the form of materials or supplies to be consumed
Five step model
- Identify the contract with the customer
- Identify the performance obligations in the contract
- Determine the transaction price
- Allocate the transaction price to the performance obligations in the contract
- Recognise revenue when (or as) a performance obligation is satisfied
Revenue: at fair value of the consideration received or receivable
Inventory: lower of costs and net realisable value
NRV:
- Estimated selling price less estimated costs of completion and selling costs
- Estimated on an item by item basis or groups of similar items
Cost:
- Of purchase
- Of conversion
- Other costs in bringing inventories to their present location and condition
Cost Formulae
FIFO
Weighted average cost

Further question practice

1 Knowledge diagnostic

Before you move on to question practice, confirm you are able to answer the following questions having studied this chapter. If not, you are advised to revisit the relevant learning from the topic indicated.

	Confirm your learning
1.	Can you recall the five steps to recognising revenue based on the requirements in IFRS 15? (Topic 2)
2.	Do you understand what is meant by a performance obligation and can you give examples of these? (Topic 2)
3.	Can you correctly deal with 'bundled contracts' where there are goods and services, and/or free to the customer elements within the contract? (Topic 2)
4.	Are you able to correctly identify which costs may be included in the valuation of inventory at the period end? (Topic 3)
5.	Can you identify issues regarding the net realisable value of items of inventory, and which accounting procedures should be followed if a write-down of inventory is required? (Topic 3)

2 Question practice

Aim to complete all self-test questions at the end of this chapter. The following self-test questions are particularly helpful to further topic understanding and guide skills application before you proceed to the next chapter.

Question name	**Learning benefit from attempting this question**
White Goods Ltd	This question tests your understanding of how to recognise revenue and finance income when a transaction price includes a significant financing component.
Quick Ltd	This question provides useful revision of the absorption of costs in inventory.
Parson plc	A series of scenarios requiring accounting adjustments to be made in respect of revenue for the entity. A short explanation answer is required for part (b) to test your understanding and ability to communicate clearly.
Scramjet	This question tests you understanding of the identification of the performance obligations in a contract and when (and how much) revenue may be recognised.

Once you have completed these self-test questions, it is beneficial to attempt the following questions from the Question Bank for this module. These questions have been selected to introduce exam-style scenarios to help you improve knowledge application and professional skills development before you start the next chapter.

Question name	Learning benefit from attempting this question
Ashgill plc	This is a preparation of financial statements question requiring a number of adjustments, including revenue and inventories as well as other adjustments you will be familiar with from your *Accounting* studies and the early chapters of the *Financial Accounting and Reporting* module. The question is for 30 marks, so allow 54 minutes to complete it.
Avebury plc (issue 1 only)	This question contains a loyalty card scheme. It requires you to apply IFRS 15 to determine the correct accounting treatment for the recognition of revenue.
Tarascon Ltd (issue 3 only)	Issue 3 includes two different revenue streams which you must determine how to account for: a pay-as-you-go membership and an annual membership for access to an entertainment venue.

Refer back to the learning in this chapter for any questions which you did not answer correctly or where the suggested solution has not provided sufficient explanation to answer all your queries. Once you have attempted this question, you can continue your studies by moving onto the next chapter.

Technical reference

You will need to be familiar with the whole of IAS 2, *Inventories* for your exam. For IFRS 15, *Revenue from Contracts with Customers*, you will need to focus on the key paragraphs stated below, however, noted exclusions include variable consideration (paras. 50-59) and incremental costs of obtaining a contract (paras. 91-104).

The paragraphs listed below are the key references you should be familiar with.

1 IFRS 15, *Revenue from Contracts with Customers*

- Identifying the contract – **IFRS 15 (9)**
- Identifying the performance obligations – **IFRS 15 (22)**
- Satisfaction of performance obligations – **IFRS 15 (31)**
- Performance obligations satisfied over time – **IFRS 15 (35)**
- Performance obligations satisfied at a point in time – **IFRS 15 (38)**
- Determining the transaction price – **IFRS 15 (47)**
- Significant financing component – **IFRS 15 (60)**
- Allocating the transaction price to the performance obligations – **IFRS 15 (73)**

2 IAS 2, *Inventories*

- Measurement and disclosure, but not recognition – **IAS 2 (1)**
- Inventories are to be measured at the lower of cost and net realisable value – **IAS 2 (9)**
- Cost = expenditure incurred in bringing the items to their present location and condition, so the cost of purchase and the cost of conversion – **IAS 2 (10)**
  - Fixed costs included by reference to normal levels of activity – **IAS 2 (13)**
- Cost formulae: FIFO or weighted average – **IAS 2 (25)**
  - Use same formula for all inventories with similar nature
- Net realisable value takes costs to complete into account, as well as selling costs – **IAS 2 (6)**
- Disclosures include accounting policies, carrying amounts and amounts recognised as an expense – **IAS 2 (36 and 38)**

Answer the following questions.

1 Agent and principal

Webber Ltd sells two types of product, the sleigh and the sled. Webber Ltd sells the sleigh as an agent of Caplin Ltd receiving commission of 15% on selling price. Webber Ltd sells the sled as principal at a gross margin of 30%. Webber Ltd has satisfied its performance obligations in respect of the sales.

The following information relates to the year ended 30 September 20X8.

	Sleighs	**Sleds**
	CU	**CU**
Total sales	200,000	75,000
Gross profit	60,000	22,500

Requirement

According to IFRS 15, *Revenue from Contracts with Customers* what amount of revenue should Webber recognise in total for sleighs and sleds for the year ended 30 September 20X8?

2 Significant financing component

On 1 January 20X0, Alex Ltd supplied goods to Sydney Ltd for an agreed sum of CU600,000. This amount becomes payable on 31 December 20X2. Sydney Ltd could have bought the goods for cash of CU450,000 on 1 January 20X0. The discount rate that would be reflected in a separate financing arrangement between Alex Ltd and Sydney Ltd at 1 January 20X0 is 10%.

Requirement

In accordance with IFRS 15, *Revenue from Contracts with Customers* what amounts for revenue and interest income should Alex Ltd record in profit or loss relating to this transaction for the year ended 31 December 20X0?

3 Southwell Ltd

Southwell Ltd, a property development company, sold a property with a carrying amount of CU4.5 million for CU5 million to Financier Ltd on 1 January 20X4 (the current market value is CU5 million). Southwell Ltd retains the right to occupy the property and obligation to maintain the building and has a forward option to repurchase the property in two years' time for CU6 million. It is expected that the market value of the property will be CU6 million in two years' time. The annual rate for 20% over two years is 9.5%.

Requirement

In accordance with IFRS 15, *Revenue from Contracts with Customers* what should be recognised in the financial statements relating to this transaction for the year ended 31 December 20X4?

4 White Goods Ltd

White Goods Ltd sells an electrical appliance for CU2,400 on 1 October 20X7 making a mark up on cost of 20%. The customer is given a two-year interest-free credit period but can use the appliance from the date of receiving the item (also 1 October 20X7). An appropriate interest rate is 9%.

In accordance with IFRS 15, *Revenue from Contracts with Customers*, what amount should the company recognise as revenue from the sale of the appliance in profit or loss for the year ended 30 September 20X8?

5 Taunton plc

Taunton plc manufactures spare parts for a range of agricultural equipment. These are sent from its Bangladeshi factory to its various distribution centres in Bangladesh.

According to IAS 2, *Inventories*, which of the following expenses should be included as part of the cost of finished goods inventories?

A Rectification costs of a lorry-load of parts that were badly damaged in an accident en route to one of the Bangladeshi distribution centres.

B Expenses paid to the firm's lorry-drivers for transporting parts from the distribution centres to customers.

C Shipping costs for drivers and lorries to the Irish distribution centre.

D Subsistence and accommodation expenses relating to the return journey from Ireland.

6 Quick Ltd

Quick Ltd absorbs its production overheads on the basis of units produced. Cost data relating to units of production for the year to 31 December 20X3 are as follows.

Material (for 10,000 units actually produced)	CU10,000
Sub-contract labour	CU20,000
Fixed production overheads	CU50,000

There are 1,000 units in inventories at the year end. Production was half the normal activity level.

Requirement

In accordance with IAS 2, *Inventories* what should be the figure for inventories in the statement of financial position on 31 December 20X3?

7 Valuation of inventories at year end

The normal selling price of an item included in year-end inventories is CU21 per unit. The item originally cost CU15 per unit, but could only be sold at the normal selling price after modifications were made after the year end at a cost of CU5 per unit. The scrap value of the item is CU11 per unit.

Requirement

Under IAS 2, *Inventories,* at what amount should the item be included in the financial statements?

8 Hunt Ltd

Hunt Ltd has prepared the following schedule in respect of two items held in inventory.

	Purchase price of raw materials	Attributable production overheads incurred	Attributable distribution overheads to be incurred	Expected selling price
	CU	CU	CU	CU
Item X	80	10	12	85
Item Z	20	5	10	40
	100	15	22	125

Requirement

According to IAS 2, *Inventories* what is the aggregate amount at which inventories of these items should be stated in the statement of financial position of Hunt Ltd?

9 Greenmore Ltd

Greenmore Ltd is making a product for a customer. The cost to date is CU35,000. Owing to a change in government regulations an additional CU12,000 will need to be spent before the product can be sold. The customer agrees to pay half of this. The initially agreed selling price was CU40,000.

Requirement

At what amount should inventories be carried in the financial statements of Greenmore Ltd according to IAS 2, *Inventories*?

10 Parson plc

Parson plc has entered into the following transactions during the year ended 31 December 20X3.

(1) On 1 October 20X3 Parson plc received CU400,000 in advance subscriptions. The subscriptions are for 20 monthly issues of a magazine published by Parson plc. Three issues of the magazine had been despatched by the year end. Each magazine is of the same value and costs approximately the same to produce. The payment in advance does not constitute a significant financing component.

(2) Parson plc made a major sale on 1 January 20X3 for a fee of CU450,000, which related to a completed sale and after-sales support for three years. The cost of providing the after-sales support is estimated at CU50,000 pa, and the mark-up on similar after-sales only contracts is 20% on cost.

(3) The food division of Parson plc allows other companies to use its brand name at various food outlets. On 1 January 20X3 a new outlet was opened, the licence paid of CU500,000 to cover the initial brand launch. The licence is for five years, and the company will pay Parson plc an additional annual fee of CU60,000 commencing on 1 January 20X3 to cover marketing, managerial and other support services provided by Parson plc during the period. Parson plc has estimated that the cost of providing these services is CU80,000 pa, and has achieved a gross margin of 20% on providing similar services on other contracts.

Requirement

Prepare extracts from Parson plc's financial statements for the year ended 31 December 20X3, clearly showing how each of the above would be reflected. Notes to the financial statements are not required.

11 Scramjet

Scramjet is a Bangladesh-based airline which has traditionally operated in the low-cost sector, but which is trying to attract a greater proportion of business travellers.

The airline sells tickets in different booking classes, each of which has a different set of booking conditions concerning changes, refunds, and so on.

The further in advance flights are paid for and the more restrictive the booking conditions (such as not allowing changes or refunds in any circumstances), the less the flights cost.

At the company's year end of 31 December 20X6, there are a large number of passengers who have paid in advance for their flights. A sample of two passengers below illustrates some of the booking conditions.

(1) On 1 December 20X6, Ali booked a one-way flight to Paris in booking class Y, at a price of CU150. This is the most flexible booking class and allows Ali to take any flight operated by

Scramjet to Paris within six months of the booking being made. At the year-end, Ali had not used the one- way ticket. These flights are flexible, but once paid for, no refunds may be given.

(2) On 7 August 20X6, Sam booked a return flight for the new year holiday from Stansted to Prague. The outward flight was on 28 December 20X6 and the return flight was on 3 January 20X7. The tickets are non-refundable and the flights cannot be changed once booked. Sam paid CU70 for the flight in each direction and has paid for the flights in full by the year end.

Note: Each flight is a separate performance obligation.

Requirement

Applying IFRS 15, *Revenue from Contracts with Customers*, explain how each of the above transactions should be reported in the financial statements of Scramjet at 31 December 20X6. State any criticisms you may have of the accounting treatment that IFRS 15 requires.

12 Latentile Ltd

Latentile Ltd is a newly formed company, which uses a chemical process to manufacture a revolutionary new roof covering, which it sells at a mark-up of 25% on cost. Its inventories consist of raw material, work in progress and finished goods, and at the end of its first year of trading it is having problems valuing inventories.

You ascertain the following information.

(1) **Raw material**

The process needs at least 100,000kg of clay to continue working, but a physical inventory count reveals that the machinery contains 108,000kg.

The original cost of the initial 100,000kg to set up the process was 30p per kg and you find an invoice to show that the last consignment of 20,000kg cost 31p per kg. All other consignments in the year (a total of 200,000kg) cost 32p per kg.

(2) **Work in progress**

The work in progress is currently all 60% complete and you discover that there are 50,000 units currently going through the process.

The total number of complete units for the period was, as anticipated, 800,000. The costs for the process for the period were as follows.

	CU
Raw materials	200,000
Direct labour	242,000
Factory overheads	191,000
Administrative expenses attributable to production	114,000
Distribution costs	90,000

(3) **Finished goods**

There were 70,000 units in inventories.

Of (1) above, it was intended to sell 20,000 units at 75p per unit, a discount of one third on normal selling price, in a future promotional campaign (a further 10p per unit distribution cost is to be incurred).

Requirements

12.1 Explain how IAS 2, *Inventories* applies the accrual and the going concern bases of accounting.

12.2 For each of the above categories of inventory, suggest a method of valuation and show the value as it would appear in the statement of financial position.

12.3 If the information regarding costs for the period were not available, suggest an alternative method of valuing finished goods.

Now go back to the Introduction and ensure that you have achieved the Learning outcomes listed for this chapter.

Answers to Interactive questions

Answer to Interactive question 1

Skyward plc is contracted to build and supply three aircraft. Each aircraft is a distinct good that is to be delivered to PEG once it has been completed and, prior to delivery, PEG cannot use or control the aircrafts. The performance obligations in this case are the delivery of each of the individual aircraft at six-month intervals.

Answer to Interactive question 2

Revenue CUnil

Revenue should be recognised when the drills are delivered to the customer. This is the point at which the performance obligation is satisfied. Until then no revenue should be recognised and the deposits are recognised as contract liabilities in the statement of financial position, to reflect the DIY store's obligation to transfer the goods to the customer at a point in the future.

Answer to Interactive question 3

(a) This is a contract with performance obligations satisfied over time and revenue is recognised dependent on progress towards satisfaction of the performance obligation. Based on costs incurred/total expected costs, 33% of the performance has been completed to date.

Revenue is therefore recognised at 33% of CU210,000 = CU69,300.

(b) The costs to complete cannot be measured reliably. Since the costs basis is used to assess progress towards satisfaction of the performance obligation, the outcome of the contract cannot be estimated reliably. Therefore revenue is recognised to the extent of the costs incurred which are recoverable ie, CU40,000.

Answer to Interactive question 4

Contract revenue CU60,000

If the outcome of a services transaction cannot be estimated reliably, revenue should only be recognised to the extent that expenses incurred are recoverable from the customer.

Answer to Interactive question 5

		CU
Revenue	– sale of goods (W2)	1,363,636
	– sale of services (W2)	22,727
Total		1,500,000

The standalone price of the after-sale support must be included. This is effectively revenue and included as if the revenue was charged at full price.

WORKINGS

(1) **Calculation of the effective value of the contract**

	CU
Standalone price of after-sale support: (Cost + extra 20% margin)(120,000/(100% – 20%))	150,000
Standalone price of goods	1,500,000
Total 'true' revenue	1,650,000

(2) **Calculation of the revenue for the period**

Revenue period is 1 January–30 April 20X7 = 4 months

Transaction price allocated to goods: 1,500,000 (contract value) / 1,650,000 (W1) × CU1.5m = CU1,363,636

Transaction price allocated to after-sale support: 150,000 / 1,650,000 × CU1.5m = CU136,364

Total contract value (1,364,000 + 136,000) = CU1,500,000

Revenue in respect of the goods which were transferred across to the customer recognised in full (CU1,363,636)

Revenue for sale of services recognised in the four months to 30 April 20X7 should be CU136,364/2 years × 4/12 = CU22,727

Total revenue for the period: 22,727 + 1,363,636 = CU1,386,363

Answer to Interactive question 6

	CU
Standalone price of after-sales support (2 × (35,000/82.5%))	84,848
Standalone price of system	800,000
Total	884,848
Transaction price allocated to the provision of the system (800,000/884,848 × CU800,000)	723,288

The transaction price allocated to the provision of the system is recognised as revenue in the year.

Answer to Interactive question 7

Before the fire, overheads were allocated to inventory at 50p per unit (CU50,000 / 100,000)

The underproduction as a result of the fire does not result in any additional inventory value and should therefore not result in an additional overhead cost per unit. The overheads should still therefore be allocated on the basis of 50p per item, leading to overheads of CU37,500 being included in inventory. The CU12,500 balance of overhead cost must be recognised as an expense in the year.

Answer to Interactive question 8

	Include in cost of inventories	Recognised as an expense as incurred
Supplier's gross price for raw materials	Yes	
Quantity discounts allowed by supplier	Yes	
Purchase taxes and duties charged by supplier and recoverable from taxing authorities	n/a, because recoverable	n/a, because recoverable
Costs of transporting materials to the business' premises	Yes	
Labour costs directly incurred in the processing of raw materials	Yes	
Variable costs, such as power, incurred in the processing of raw materials	Yes	
Fixed production costs/overheads, such as rent for the processing factory and depreciation charges on the plant used in the processing	Yes, but see commentary in text	
Costs of holding finished goods in inventory		Yes
Costs of transporting goods to customer on sale		Yes
Purchase taxes charged to customer on sale	n/a as not an expense – it is part of the sales price	n/a as not an expense – it is part of the sales price
Commission payable to salesmen on sale of the goods		Yes
Allowance for irrecoverable debts in relation to trade receivables		Yes
Costs of accounts department		Yes
Head office costs relating to the overall management of the business		Yes

Answer to Interactive question 9

Department A: Selling price of inventories CU30,000 less gross profit 25% = CU22,500

Department B: If selling price is cost plus 50%, then selling price must be 150% of cost and the gross profit margin must be 50/150 = 33.3%. Selling price of inventories CU21,000 less gross profit

33.3% = CU14,000.

Answers to Self-test questions

1 Agent and principal

	CU
Revenue recognised as agent (CU200,000 × 15%)	30,000
Revenue recognised as principal	75,000
Total revenue	105,000

2 Significant financing component

At the time of supply, revenue is recognised based on the fair value of the consideration receivable, which is the cash price of CU450,000. Interest will then be accrued until payment is made. For the year ended 31 December 20X0 the interest charge (finance income) is CU450,000 × 10% = CU45,000. The receivable at 31 December 20X0 would be CU495,000 (CU450,000 + CU45,000).

3 Southwell Ltd

There is a forward option for Southwell to repurchase the property and therefore Financier Ltd is limited in its ability to direct the use of the property or obtain the remaining economic benefits from it. Therefore, the substance of the transaction is not a sale.

The repurchase price is more that the selling price. Therefore, this is treated as a financing arrangement; Southwell should not derecognise the property and should recognise the proceeds from Financier as a financial liability.

Initial financing:

	CU000	CU000
DR Cash	5,000	
CR Loan		5,000

Interest:

	CU000	CU000
DR Interest (Profit or loss) (5m × 9.5%)	475	
CR Loan		475

Total loan liability at 31 December 20X4 is CU5.475 million.

4 White Goods Ltd

Due to the credit period given to the customer, this is deemed to be a transaction with a significant financing component. Revenue is recognised at the fair value of the consideration receivable. The amount receivable is discounted to present value = CU2,400 × $1/1.09^2$ = CU2,020.

CU2,020 is recognised as revenue on 1 October 20X7, which is when the customer obtains control of the appliance.

Subsequently the CU380 difference between the transaction price of CU2,400 and the discounted price of CU2,020 is recognised as a finance income. In the year ended 30 September 20X8 finance income of CU2,020 × 9% = CU182 is recognised in profit or loss and accrued to the outstanding receivable.

5 Taunton plc

The correct answer is:

C Shipping costs for drivers and lorries to the Irish distribution centre.

Cost comprises all costs of purchase, conversion and other costs incurred in bringing the inventories to their present location and condition (IAS 2: para. 10). Only C meets this definition of costs.
Abnormal costs such as A are effectively excluded by IAS 2 paragraph 16(a).

6 Quick Ltd

		CU		CU
Material	$\frac{CU10{,}000}{10{,}000} =$	1.0		
Labour	$\frac{CU20{,}000}{10{,}000} =$	2.0		
Overheads*	$\frac{CU50{,}000}{(10{,}000 \div 50\%)} =$	2.5		
		5.5	× 1,000	5,500

7 Valuation of inventories at year end

Inventories should be measured at the lower of cost and NRV. Cost = CU15

NRV = (21 – 5) = CU16

The inventories should be carried at CU15 per unit.

8 Hunt Ltd

Inventories should be measured at the lower of cost and NRV.

	Cost	NRV	Lower
	CU	CU	CU
Item X	90	73	73
Item Z	25	30	25
Total			98

9 Greenmore Ltd

Cost = CU35,000

NRV = 40,000 – 12,000 costs to complete + 6,000 customer contribution = CU34,000

Inventories should therefore be carried at CU34,000.

10 Parson plc

Statement of financial position as at 31 December 20X3

	CU
EQUITY AND LIABILITIES	
Non-current liabilities	
Contract liability (W2)	280,000
Current liabilities	
Contract liability (W2)	340,000

Statement of profit or loss for the year ended 31 December 20X3

	CU
Revenue (W1)	790,000

WORKINGS

(1) **Revenue**

	CU
Transaction (1) (3/20 × 400,000)	60,000
Transaction (2)	
Sale (450,000 – (50,000 × 120% × 3))	270,000
After-sales support Year 1 (50,000 × 120%)	60,000
Transaction (3)	
Initial fee (500,000 – (40,000 (W2) × 5))	300,000
Continuing fee Year 1 (80,000 × 100/80)	100,000
	790,000

(2) **Contract liability**

	Current	Non-current
	CU	CU
Transaction (1) (400,000 × 12/20, 5/20)	240,000	100,000
Transaction (2) (50,000 × 120% for Years 2 and 3)	60,000	60,000
Transaction (3) (100,000 – 60,000 for Years 2 to 5)	40,000	120,000
	340,000	280,000

C H A P T E R 6

11 Scramjet

IFRS 15 requires that revenue is recognised by reference to the satisfaction of performance obligations at the end of the reporting period. This means that Scramjet may have taken bookings which are entirely non-refundable but the revenue associated with payments for tickets should be held in the statement of financial position as a contract liability until either the flight is taken or the ticket expires. In both cases, the tickets are non-refundable, so there is certainty that the revenue can be recognised, albeit with different dates as stated below:

Ali

Ali's flight has not been taken at the year-end and so the CU150 received should also be recognised as a contract liability in the statement of financial position at 31 December 20X6. The flights are flexible, however, they are still non-refundable, so it is correct to recognise the contract liability, however, the revenue may not be recognised until the flight is taken.

Sam

Under IFRS 15, the revenue for the outbound flight should be recognised as that part of the performance obligation has been satisfied prior to the year end: CU70 should be recognised in profit or loss and CU70 should remain in the statement of financial position as a contract liability until Sam's flight back is completed on 3 January 20X7.

12 Latentile Ltd

12.1 Accrual basis of accounting

The cost of unsold or unconsumed inventories is incurred in the expectation of future economic benefits. When such benefits will not arise until a subsequent accounting period, the related costs should be carried forward and matched with the revenue when it arises. The recognition of year-end inventories achieves this carry forward.

Going concern basis of accounting

The very act of recognising closing inventories as assets at the year-end implies that the business intends to continue in operational existence for the foreseeable future.

If the company were to cease trading, inventory would be measured at the lower of cost or net realizable value, which may be nil (if the company is no longer actively selling inventory, then it should be written off).

12.2 Suggested methods of valuing inventories

Given the limited information the following methods would be appropriate in the circumstances.

Raw material

IAS 2 allows either a first in, first out (FIFO) formula or a weighted average cost (WAC) formula.

Given the fact that clay is presumably continually added to the machinery, WAC would seem the most appropriate basis.

	CU
100,000kg @ 30p	30,000
200,000kg @ 32p	64,000
20,000kg @ 31p	6,200
320,000	100,200

= 31.3p per kg

Closing raw materials would therefore be measured at CU33,804 (108,000 × 31.3p).

Work in progress

This could be measured using a weighted average cost, given that total cost and total output are known.

Total output (800,000 + (60% × 50,000))	830,000 units
Total costs (excluding distribution costs)	CU747,000
Thus average cost per unit = 90p	
Carrying amount of WIP (50,000 × 60% × 90p)	CU27,000

Finished goods

Again, a weighted cost could be used of 90p per unit. This would be applicable to 50,000 units, with the remaining 20,000 units being measured at net realisable value of 65p (75p – 10p).

	CU
50,000 at 90p	45,000
20,000 at 65p	13,000
	58,000

Thus inventories would appear as follows.

	CU
Raw material	33,804
Work in progress	27,000
Finished goods	58,000
	118,804

12.3 Alternative valuation method for finished goods

If details regarding total costs were not known, adjusted selling price could be used since the cost structure is known.

	p
Normal selling price (75p discounted price × 3/2)	112.5
Less gross profit (112.5 × 25/125)	(22.5)
Cost re 50,000	90.0

Thus finished goods inventories would be measured as before.

IAS 2 allows the above practice, used by the retail industry, on the basis that the result can be a very close approximation to cost.